Observer Frames and Relativity from Event-Order Geometry

Lee Smart

Institute of Vibrational Field Dynamics

contact@vibrationalfielddynamics.org

@vfd_org

April 2026

Status: Preprint (not peer-reviewed)

Abstract

Paper XXIII derived a strict partial order on admissible overlap events between two phase-rotated 600-cell copies, interpreting the ordering as an emergent temporal structure. The present paper shows that this event poset carries observer-dependent structure analogous to key kinematic features of special relativity.

An observer is defined as a linear extension of the event partial order equipped with a sampling measure over events. Different observers assign different time functions to the same event set. The paper proves: (1) relativity of simultaneity — incomparable events can be ordered differently by different observers; (2) that differing observer measures yield non-unique elapsed-time assignments along shared event chains, providing a time-dilation analogue; (3) worldlines arise as maximal chains in the poset. A worked example on the dual-pentagon system demonstrates all constructions explicitly.

The paper does not derive Lorentz invariance, the metric tensor, or the Einstein equations. It establishes that the kinematic phenomena of frame-dependent time and simultaneity emerge from the event-order geometry without assuming spacetime.

1 Introduction

Paper XXIII [Smart, 2026] established that the phase-overlap geometry of two conjugate 600-cell copies produces a strict partial order $\prec$ on an event set $\mathcal{E}$, with no canonical global clock. The present paper converts this mathematical structure into observer-dependent physics.

The central question is: given a partial order on events with no preferred total ordering, how do different observers reconstruct temporal experience, and do the resulting differences reproduce features of relativistic kinematics?

Scope

This paper derives observer-dependent time functions, proves relativity of simultaneity, and exhibits a time-dilation analogue. It does not derive Lorentz invariance, construct a metric, or claim equivalence with special relativity. It establishes that relativity-like kinematic structure emerges from the event poset without spacetime assumptions.

2 Observer Definition

Definition 1 (Observer)**.** *An observer on the event poset $(\mathcal{E}, \prec)$ is a pair*

$$O = (\leq_O,\ \mu_O) \tag{1}$$

where:

- *$\leq_O$ is a linear extension of $\prec$: a total order on $\mathcal{E}$ such that $e_1 \prec e_2 \Rightarrow e_1 \leq_O e_2$,*
- *$\mu_O : \mathcal{E} \to \mathbb{R}_{>0}$ is a sampling measure: a positive function assigning a "duration weight" to each event.*

Remark 2. *The linear extension $\leq_O$ determines what order the observer assigns to events. The sampling measure μ_O determines how much time the observer associates with each event. Together they produce a complete observer time function.*

Remark 3. *No physical velocity, mass, or energy is assumed. The sampling measure is a structural degree of freedom intrinsic to the observer's relationship with the event poset.*

3 Observer Time Functions

Definition 4 (Observer time)**.** *For an observer $O = (\leq_O, \mu_O)$, the observer time function is:*

$$t_O(e) = \sum_{\substack{e' \in \mathcal{E} \\ e' \leq_O e}} \mu_O(e') \tag{2}$$

That is, the time assigned to event e is the cumulative measure of all events up to and including e in the observer's ordering.

Proposition 5 (Order preservation)**.** *For any observer O:*

$$e_1 \prec e_2 \ \Rightarrow\ t_O(e_1) < t_O(e_2) \tag{3}$$

Proof. If $e_1 \prec e_2$, then $e_1 \leq_O e_2$ (since $\leq_O$ extends $\prec$) and $e_1 \neq e_2$. The sum defining $t_O(e_2)$ includes all terms in $t_O(e_1)$ plus at least $\mu_O(e_2) > 0$. Therefore $t_O(e_2) > t_O(e_1)$. □

Remark 6. *The time function t_O is not unique: it depends on both the choice of linear extension and the choice of sampling measure. Different observers generically assign different times to the same events, even when they agree on the partial order.*

4 Relativity of Simultaneity

Theorem 7 (Relativity of simultaneity)**.** *If events $e_1, e_2 \in \mathcal{E}$ are incomparable under $\prec$ (neither $e_1 \prec e_2$ nor $e_2 \prec e_1$), then there exist observers O_1, O_2 such that:*

$$t_{O_1}(e_1) < t_{O_1}(e_2) \qquad \textit{and} \qquad t_{O_2}(e_2) < t_{O_2}(e_1) \tag{4}$$

Proof. Since e_1 and e_2 are incomparable, there exist linear extensions $\leq_1$ placing e_1 before e_2, and $\leq_2$ placing e_2 before e_1. (This follows from the standard extension property of finite partial orders: for any two incomparable elements, linear extensions exist placing them in either relative order.) Choose any positive measure $\mu > 0$ and set $O_1 = (\leq_1, \mu)$, $O_2 = (\leq_2, \mu)$. Then $t_{O_1}(e_1) < t_{O_1}(e_2)$ and $t_{O_2}(e_2) < t_{O_2}(e_1)$ by construction. □

Remark 8. *This is the geometric analogue of the relativity of simultaneity in special relativity: events that are spacelike-separated (incomparable in the causal order) can be ordered differently by different inertial frames (linear extensions). The VFD framework derives this from the event poset; special relativity derives it from Lorentz invariance. The structural role is analogous.*

5 Time Dilation Analogue

Definition 9 (Elapsed time along a chain)**.** *For a chain $C = (e_1 \prec e_2 \prec \cdots \prec e_n) \subset \mathcal{E}$ and observer O, the elapsed time is:*

$$\Delta t_O(C) = t_O(e_n) - t_O(e_1) = \sum_{\substack{e' \in \mathcal{E} \\ e_1 <_O e' \leq_O e_n}} \mu_O(e') \tag{5}$$

Theorem 10 (Time-dilation analogue)**.** *Let $O_1 = (\leq, \mu_1)$ and $O_2 = (\leq, \mu_2)$ be observers sharing the same linear extension but with different sampling measures. If μ_1 and μ_2 differ on the set of events lying between e_1 and e_n in the shared ordering, then:*

$$\Delta t_{O_1}(C) \neq \Delta t_{O_2}(C) \tag{6}$$

for the chain $C = (e_1 \prec \cdots \prec e_n)$. Both observers agree on the ordering of events but disagree on the elapsed time.

Proof. Since O_1 and O_2 share the ordering $\leq$, the set S of events between e_1 and e_n is identical for both. By hypothesis, μ_1 and μ_2 differ on S, so:

$$\sum_{e' \in S} \mu_1(e') \neq \sum_{e' \in S} \mu_2(e') \tag{7}$$

giving $\Delta t_{O_1}(C) \neq \Delta t_{O_2}(C)$. □

Remark 11. *This is the event-order analogue of time dilation: two observers traversing the same sequence of events measure different elapsed times. In special relativity, this arises from relative velocity; here it arises from different sampling measures. The structural effect — same events, different durations — is the same.*

Remark 12. *The sampling measure μ_O is not yet connected to physical velocity or energy. Establishing that connection requires metric structure (Paper XXV). The present paper demonstrates only that the kinematic possibility of time dilation exists in the event-order framework.*

6 Worldlines

Definition 13 (Worldline)**.** *A worldline is a maximal chain in $(\mathcal{E}, \prec)$: a totally ordered subset $W \subset \mathcal{E}$ that cannot be extended by inclusion.*

Remark 14. *A worldline is a complete sequence of events each following the previous in the partial order. Different worldlines may share some events (intersect) or be entirely disjoint.*

Definition 15 (Chain time)**.** *For an observer O and a chain $W = (e_1 \prec \cdots \prec e_n)$, the chain time (or worldline time) is:*

$$\tau_O(W) = \sum_{k=1}^{n} \mu_O(e_k) \tag{8}$$

This is the cumulative measure restricted to events on the chain itself.

Remark 16. *Chain time $\tau_O(W)$ differs in general from the observer elapsed time $\Delta t_O(W) = t_O(e_n) - t_O(e_1)$, since Δt_O includes contributions from all events between e_1 and e_n in the observer's total ordering, not just those on the chain. The chain time τ_O is the intrinsic duration along the worldline; Δt_O is the global clock reading.*

7 Relative Velocity (Proto-Concept)

A full notion of velocity requires spatial structure, which this paper does not construct. However, a proto-concept can be defined.

Definition 17 (Relative sampling rate)**.** *For two observers O_1, O_2, their relative sampling rate along a chain C is:*

$$\rho(O_1, O_2; C) = \frac{\tau_{O_1}(C)}{\tau_{O_2}(C)} = \frac{\sum_{e \in C} \mu_1(e)}{\sum_{e \in C} \mu_2(e)} \tag{9}$$

When $\rho \neq 1$, the two observers experience the same chain at different rates. This is the structural precursor to relative velocity: two observers "moving" through the event structure at different "speeds" (sampling rates).

Remark 18. *This proto-velocity does not have units of distance/time. It is a dimensionless ratio of sampling measures. Connecting it to physical velocity requires the spatial metric structure of Paper XXV.*

8 Worked Example: Dual Pentagons

Example 19 (Extending the pentagon system)**.** *Using the dual-pentagon event set from Paper XXIII: 5 events at $\theta_b = 0$ and 5 events at $\theta_b = 2\pi/5$.*

Two observers. *Define:*

- *$O_1 = (\leq_1, \mu_1)$: orders the $\theta_b = 0$ events as $(0,0) < (1,1) < (2,2) < (3,3) < (4,4)$, with uniform measure $\mu_1(e) = 1$ for all e.*

- $O_2 = (\leq_2, \mu_2)$: *orders the* $\theta_b = 0$ *events as* $(2,2) < (4,4) < (1,1) < (3,3) < (0,0)$ *(a different permutation), with non-uniform measure* $\mu_2(e) = 2$ *for* $\theta_b = 0$ *events and* $\mu_2(e) = 1$ *for* $\theta_b = 2\pi/5$ *events.*

Both observers agree: all $\theta_b = 0$ *events precede all* $\theta_b = 2\pi/5$ *events.*

Simultaneity. O_1 *assigns* $t_{O_1}(0,0) = 1 < t_{O_1}(1,1) = 2$. *But* O_2 *assigns* $t_{O_2}(0,0) = 10 > t_{O_2}(1,1) = 6$. *The events* $(0,0)$ *and* $(1,1)$ *are ordered differently by the two observers: relativity of simultaneity demonstrated.*

Time dilation. *Consider the chain* C: *any* $\theta_b = 0$ *event followed by any* $\theta_b = 2\pi/5$ *event (e.g.* $(0,0) \prec (0,4)$*).*

- O_1: $\Delta t_{O_1}(C) = t_{O_1}(0,4) - t_{O_1}(0,0)$. *With uniform* $\mu_1 = 1$, *if* $(0,0)$ *is the 1st event and* $(0,4)$ *is the 6th:* $\Delta t = 5$.

- O_2: *with* $\mu_2 = 2$ *for the first layer, if* $(0,0)$ *is the 5th event in layer 1:* $t_{O_2}(0,0) = 10$. *Then* $(0,4)$ *in layer 2 at position 6:* $t_{O_2}(0,4) = 10 + 1 = 11$. *So* $\Delta t = 1$.

Same chain, different elapsed times: $\Delta t_{O_1} = 5 \neq \Delta t_{O_2} = 1$. *Time dilation analogue demonstrated.*

Relative sampling rate. $\rho(O_1, O_2; C) = 5/1 = 5$. *Observer* O_1 *accumulates 5 times more elapsed time than* O_2 *along this chain.*

9 Relation to Special Relativity

Table 1: Structural comparison: VFD event-order framework vs. special relativity.

Feature	VFD (this paper)	Special relativity
Events	Admissible vertex pairs	Spacetime events
Ordering	Birth-angle partial order	Causal structure
Observer	Linear extension + measure	Inertial frame
Time function	Cumulative sampling measure	Proper time
Simultaneity	Observer-dependent (proved)	Frame-dependent
Time dilation	Sampling-rate differences (exhibited)	Velocity-dependent
Worldline	Maximal chain in poset	Timelike curve
Velocity	Proto-concept (sampling ratio)	dx/dt

Remark 20 (Epistemic status)**.** *The structural analogies in Table 1 are exhibited, not derived from Lorentz invariance. The VFD framework produces the same qualitative phenomena (observer-dependent simultaneity, time dilation, worldline structure) from a different starting point (event poset with sampling measures). Whether this structural analogy can be promoted to full Lorentz covariance requires metric structure not yet available.*

10 Discussion

10.1 What is established

- Observer defined as linear extension + sampling measure (Definition 1).

- Observer time function constructed explicitly (Definition 4, Proposition 5).
- Relativity of simultaneity proved (Theorem 7).
- Time-dilation analogue exhibited under differing observer measures (Theorem 10).
- Worldlines defined as maximal chains (Definition 13).
- Worked example demonstrates all constructions.

10.2 What is not established

- No Lorentz invariance. The framework has discrete (H_4) symmetry, not continuous Lorentz symmetry.
- No metric. The sampling measure gives duration differences but not distances.
- No Einstein equations. The connection between observer structure and gravitational dynamics is not addressed.
- No physical velocity. The relative sampling rate is a structural precursor, not a derived physical quantity.
- The sampling measure μ_O is a structural degree of freedom, not yet connected to physical observables (mass, energy, motion).

10.3 Programme position

- Paper XXIII: event structure and emergent temporal ordering.
- Paper XXIV (this paper): observer frames and relativity-like kinematics.
- Paper XXV (future): metric structure and the route to dynamics.

11 Conclusion

The event poset of Paper XXIII, equipped with observer-dependent sampling measures, supports a kinematic precursor of relativistic structure: observer-dependent simultaneity, time-dilation analogues, and worldline geometry. These features are proved, not assumed. No spacetime manifold, metric, or Lorentz symmetry is postulated.

The key structural insight is that the non-uniqueness of the global clock (Paper XXIII) is not a deficiency but a feature: it is the geometric origin of observer dependence. Different observers are different ways of completing the partial order into a total order, and different sampling measures produce different experienced durations.

The next step is to equip the event structure with spatial and metric content (Paper XXV), connecting the sampling measure to physical observables and determining whether the full Lorentz-covariant structure can be recovered from the 600-cell geometry.

References

Lee Smart. Event structure and the emergence of time from phase-overlap geometry. 2026. Paper XXIII. Preprint, Institute of Vibrational Field Dynamics.